23CSE306

MACHINE LEARNING

MODEL ANSWER BOOKLET

(handwritten sample)

Student name :_____

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Complete study version

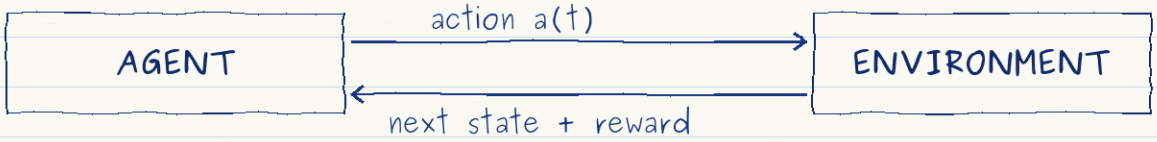
Answers to Q1–Q10, including both OR alternatives

Prepared from the supplied final question paper

Q1 – Reinforcement Learning and data preprocessing

1) Reinforcement Learning (RL)

Reinforcement Learning is a learning framework in which an agent learns by interacting with an environment. At time t, the agent observes a state s(t), chooses an action a(t), receives a numerical reward r(t+1), and moves to the next state s(t+1). The objective is not merely to obtain the largest immediate reward; it is to learn a policy that maximizes the expected cumulative return over time.



- Agent: the decision maker. In the car example, it is the driving controller.
- Environment: everything with which the *agent* interacts, including the road and traffic.
- State: the information used for a decision, such as speed, lane position and nearby objects.
- Action: a control command such as steering, throttle, braking or a lane-change decision.
- Reward: scalar feedback that says whether the recent behaviour was *desirable*.
- Policy: a rule *pi(a* given s) that maps a state to an action or an action probability.
- Return: G(t) = r(t+1) + gamma*r(t+2) + gamma^2*r(t+3) + ... , where gamma *discounts distant* rewards.
- Value functions estimate long-term return. Q(s,a) measures the value of taking action a in state s.

Q1 (cont.)

Learning requires a balance *between* exploration, which tries unfamiliar actions to collect information, and exploitation, which uses the best action currently known. Common approaches include Q-learning and Deep Q-Networks for *action* values, policy-gradient methods for directly improving a policy, and actor-critic methods that *learn* both a policy and a value estimate. Training may be organized into episodes, with terminal states such as reaching the destination or ending a simulation.

2) Autonomous-car environment

For an autonomous car, the environment contains the lane geometry, curves, intersections, traffic signs and lights, other vehicles, pedestrians, cyclists, obstacles, weather, road friction and the destination map. The observable *state* can be constructed *from* camera, LiDAR, radar, GPS and inertial sensors. Useful state variables include vehicle speed, heading, lane offset, *distance* and relative speed to nearby objects, signal status, free-space map and route progress.

The action space may be continuous, such as steering angle, throttle and brake pressure, or partly discrete, such as keep lane, change lane, stop and turn. Because real-world *exploration* is unsafe, most learning should first occur in a high-fidelity simulator, followed by carefully constrained testing and safety monitoring.

3) Reward design for the car

- Give a strong positive *reward* for safely reaching the destination.
- Give small positive rewards for forward route progress, *correct* lane position and *maintaining* a safe following distance.
- Reward obeying traffic lights, speed limits and right-of-way *rules*.
- Reward smooth steering, acceleration and braking so that passenger comfort and energy use are considered.
- Give a very large negative reward for a collision. Also penalize a near miss, *lane* departure, red-light violation, speeding, excessive delay and harsh control actions.
- Use terminal conditions for collision, *an* unrecoverable off-road state, or successful arrival.

Q1 (cont.)

Reward shaping *must* be designed carefully. If progress is rewarded without safety constraints, the car may learn to drive quickly but dangerously. A practical system therefore combines the *learned* objective with hard safety rules, constrained action limits and independent emergency braking.

4) Why preprocessing and feature transformation are essential

A machine-learning algorithm learns the patterns present in its input. *If* the input *contains* errors, *incompatible* scales or leakage, the model learns misleading patterns. *Proper preprocessing* makes the representation consistent with the assumptions of the algorithm and *improves* numerical stability, training speed and *generalization*.

- Clean missing values, impossible values, duplicates and inconsistent units. Missingness can be imputed and, when useful, represented by a missing-value indicator.
- Treat outliers using investigation, clipping, robust scaling or suitable transformations. A few *extreme* values *can* dominate a loss or a distance measure.
- Encode categorical variables correctly. One-hot, ordinal or target-based encodings should be chosen according to meaning and fitted without leaking the test labels.
- Scale numerical features. Standardization or normalization is crucial for KNN, K-Means, SVMs and *gradient-based* models because a large-unit feature can otherwise dominate.
- Transform skewed variables using log or power transformations. This can stabilize variance and make relationships easier for a model to learn.
- Construct meaningful features, remove irrelevant or duplicated variables, and reduce very high dimensionality when needed.
- Fit every learned transformation only on the *training* set, then apply exactly the same pipeline to validation, test and production data.

> *Correct order: split data -> fit preprocessing on training data -> transform all splits -> train and evaluate.*

Thus preprocessing is not cosmetic. It affects the geometry of the feature space, the conditioning of optimization, the validity of evaluation and finally the reliability of the trained model.

Q2 – OR alternative: customer segmentation and accuracy

1) Unsupervised customer segmentation

Because the company has no explicit labels such as a preferred product class or a satisfaction score, the task is to discover structure rather than predict a known target. A suitable solution is clustering, supported by customer-level feature engineering.

- Create one row per customer over a fixed time window.
- Use browsing features such as number of sessions, products viewed, category-view proportions, search count, dwell time and time since the last visit.
- Use purchase features such as recency, purchase frequency, monetary value, average basket size, categories purchased, repeat-purchase rate and return rate.
- Add behavioural ratios such as viewed-to-purchased conversion, cart-abandonment rate, discount usage and mobile-versus-desktop share.

Counts and monetary values are usually skewed, so log transformation can be applied. Missing values are handled, categorical information is encoded, and numerical features are standardized so that one high-magnitude variable does not control the distance. PCA or a learned embedding may be used to compress a very wide feature matrix.

K-Means is a simple starting method when roughly compact clusters are expected. Gaussian Mixture Models provide soft membership probabilities; hierarchical clustering gives a useful dendrogram; and DBSCAN can detect irregular dense groups and noise. The number of clusters can be selected using the elbow curve, silhouette score, stability across resamples and, most importantly, business interpretability.

Example segments and insights

- Loyal high-value customers: frequent purchases, high spend and short recency. They suit loyalty rewards, early access and retention attention.
- Active browsers with low conversion: many views and long sessions but few purchases. They reveal friction in price, trust, delivery or checkout.

Q2 (cont.)

- Bargain seekers: strong response to coupons and sale periods. They help plan promotion *timing* without giving unnecessary discounts to everyone.
- Category specialists: repeated interest in one category. They support focused recommendations, content and inventory decisions.
- New or infrequent customers: little history and long recency. Separate onboarding from re-engagement, and avoid overconfident personalization.

Unsupervised learning is appropriate because it can expose latent behavioural patterns without labelled outcomes. The company should profile every cluster, test whether it remains stable over time, and run controlled campaigns to verify that *the* segments are useful rather than merely mathematically separated.

2) Accuracy for the cat-dog model

There are 100 images. The model correctly classifies all 80 cats *and* correctly classifies none of the 20 dogs.

Accuracy = number correct / total = 80 / 100 = 0.80 = 80%.

Actual class	Predicted cat	Predicted dog
Cat (80)	80	0
Dog (20)	20	0

The result is not effective even though 80% appears high. The classes are *imbalanced*, and *a trivial* model that always says cat also obtains 80%. Dog recall is 0%, cat recall is 100%, and balanced accuracy is (100% + 0%)/2 = 50%. The confusion matrix and per-class precision, recall *and* F1 score are therefore necessary.

Q3 – Least-squares line and Decision Tree versus KNN

A) Least-squares estimate

Hours studied X	Test score Y
1	50
2	55
3	65

For the line Y = a + bX, use the *least-squares* formulas. Here n = 3, sum X = 6, sum Y = 170, sum XY = 355 and sum X^2 = 14.

$$b = [n*\text{sum}(XY) - \text{sum}(X)*\text{sum}(Y)] / [n*\text{sum}(X^2) - (\text{sum } X)^2]$$

$$b = [3*355 - 6*170] / [3*14 - 6^2] = 45 / 6 = 7.5$$

$$a = [\text{sum}(Y) - b*\text{sum}(X)] / n = [170 - 7.5*6] / 3 = 41.6667$$

Therefore, the best-fit line is Y = 41.67 + 7.50 X.

Interpretation: within *this* small *data* set, one additional hour of study is associated with an estimated increase of 7.5 marks. The fitted scores for X = 1, 2 and 3 are approximately 49.17, 56.67 and 64.17.

B) When to choose a Decision Tree over KNN

- Choose a Decision Tree when an interpretable set of if-then rules is important. A path from root to leaf gives a direct explanation.
- Trees naturally learn nonlinear thresholds and feature interactions. They can use numerical and suitably encoded categorical features.
- A tree does not require feature scaling because splits depend on ordering rather than Euclidean distance. KNN is very sensitive to scale.
- Prediction with a *trained* tree is fast and requires only a sequence of comparisons. KNN must *store* the training set and calculate many distances for each new case.
- Trees are often preferable when there are irrelevant *dimensions* or when neighbourhood distance is not meaningful. KNN suffers *in* high-dimensional *spaces*.
- KNN may be preferred for a small, clean, low-dimensional *data* set with a smooth local decision boundary *and* enough memory for instance storage.

Q3 (cont.)

A single tree can overfit and be unstable, so depth limits, minimum leaf size or *pruning* should be used. If predictive accuracy is the main goal, a tree ensemble such as Random Forest may be stronger than one tree.

Q4 – OR alternative: ensembles and Logistic Regression

1) Why use an ensemble instead of one supervised model?

An ensemble combines predictions from several models. Its main advantage is that different models make different errors. If those errors are not perfectly correlated, combining the models can produce a more stable and accurate predictor than relying on one fitted hypothesis.

- Bagging trains models on resampled data and averages them. It mainly reduces variance. Random Forest also randomizes the features considered by each tree, increasing diversity.
- Boosting trains weak learners sequentially, giving later learners more attention to residual errors. It can reduce bias and build a strong nonlinear predictor.
- Voting combines class labels or probabilities from several models.
- Stacking learns a meta-model that decides how to combine heterogeneous base-model outputs.
- Averaging reduces sensitivity to a particular sample, initialization or noisy feature. It often gives smoother probabilities and better generalization.
- Different inductive biases can complement one another: for example, a linear model captures a broad trend while a tree captures interactions.

The benefit is greatest when the component models are individually competent and sufficiently diverse. Combining many almost identical, biased models adds little. Ensembles also increase training cost, memory use and deployment complexity, and they are usually less interpretable. Validation predictions must be produced without data leakage, especially in stacking.

2) Logistic Regression in real-life applications

Logistic Regression is a probabilistic classification model. For binary output, it models the probability of class 1 using the sigmoid function:

$$p(y=1 \text{ given } x) = 1 / [1 + \exp(-(b0 + b1*x1 + \ldots + bp*xp))]$$

Equivalently, the log-odds are a linear function of the features. Parameters are fitted by maximum likelihood, commonly by minimizing cross-entropy loss. L1 or L2 regularization controls complexity and can reduce overfitting.

- Credit risk: probability that a borrower will default.

Q4 (cont.)

- Health care: probability of disease, readmission or treatment response.
- Customer analytics: churn, subscription renewal or purchase conversion.
- Email and security: spam, fraud or suspicious-transaction classification.
- Operations: probability of equipment failure or late delivery.

Strengths

- Fast to train and predict, even for large sparse data sets.
- Outputs a probability, allowing a threshold to be chosen according to the cost of false positives and false negatives.
- Coefficients are interpretable: exp(bj) is the multiplicative change in *odds* for a one-unit increase in feature j, holding other variables fixed.
- Regularization handles many features and provides a strong, transparent baseline.

Limitations and responsible use

- The decision boundary is linear in the supplied feature representation. Curves and interactions require *transformed* or interaction features.
- Strong multicollinearity makes individual coefficients *unstable*. Scaling, regularization and feature review are useful.
- Outliers, missing data and class imbalance must be handled. Accuracy alone may be misleading.
- A *coefficient* describes association under the fitted model; it does not by itself prove causation.
- Probabilities and thresholds should be checked on validation data using precision, recall, ROC-AUC, calibration and application-specific costs.

For more than two classes, one-versus-rest logistic models or *multinomial* softmax regression can be *used*. In practice, Logistic Regression is valuable *when* interpretability, probability estimates and efficient *deployment* matter.

Q5 — Preprocessing for clustering and one K-Means iteration

1) Impact of preprocessing on clustering

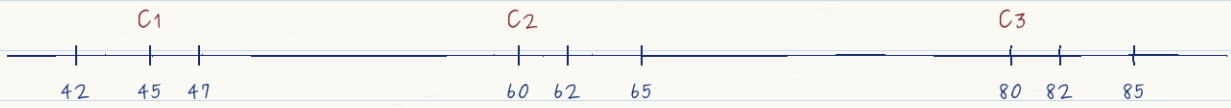
Clustering is driven directly by similarities or distances, so preprocessing can change the clusters more than the choice of algorithm. There is no *target* label to correct a poor representation.

- Scaling: with Euclidean distance, a feature measured in thousands dominates a feature measured between 0 and 1. Standardization, min-max scaling or robust scaling gives intended importance.
- *Missing* values: distance cannot be interpreted consistently when values are absent. Imputation and missingness indicators may be needed.
- Outliers: K-Means centroids are *means* and can be pulled far from the main group. Detect, cap, transform or use robust *methods* when justified.
- Categorical variables: arbitrary integer codes create false distances. Use suitable encodings or algorithms *designed* for mixed data.
- Skewness: log or power transformations reduce the dominance of long-tailed counts and spending values.
- Feature selection and dimensionality reduction: irrelevant dimensions hide neighbourhood structure. PCA can summarize correlated numerical features.
- Duplicates and sampling bias: repeated records or an unrepresentative sample can create artificial dense regions.

The entire preprocessing pipeline should be documented and fitted consistently. Cluster quality should then be checked by internal measures, stability and real-world *interpretability*.

2) One iteration of K-Means for the weights

Initial centroids are $C1 = 45$, $C2 = 60$ and $C3 = 80$. *Each* one-dimensional weight is assigned to the closest centroid *using* absolute distance, which is Euclidean distance in one dimension.



Q5 (cont.)

Cluster	Assigned weights after first assignment	Updated centroid
C1	42, 45, 47	(42+45+47)/3 = 44.67
C2	60, 62, 65	(60+62+65)/3 = 62.33
C3	80, 82, 85	(80+82+85)/3 = 82.33

After one iteration: C1 = 44.67, C2 = 62.33, C3 = 82.33.

Q6 — OR alternative: two-cluster K-Means and system design

1) K-Means for {2, 4, 10, 12, 3, 20, 30, 11, 25}

For one-dimensional data, assign each value to the nearest centroid and then replace each centroid by the mean of its assigned values.

Iteration 1: start $C_1 = 4$, $C_2 = 11$

- Cluster 1 = {2, 4, 3}; new $C_1 = (2+4+3)/3 = 3$.
- Cluster 2 = {10, 12, 20, 30, 11, 25}; new $C_2 = 108/6 = 18$.

Iteration 2: centroids 3 and 18

- Cluster 1 = {2, 4, 10, 3}; new $C_1 = 19/4 = 4.75$.
- Cluster 2 = {12, 20, 30, 11, 25}; new $C_2 = 98/5 = 19.6$.

Iteration 3: centroids 4.75 and 19.6

- Cluster 1 = {2, 4, 10, 12, 3, 11}; new $C_1 = 42/6 = 7$.
- Cluster 2 = {20, 30, 25}; new $C_2 = 75/3 = 25$.

Iteration 4: centroids 7 and 25

The assignments remain unchanged, so the algorithm has converged.

Final C_1 = {2, 3, 4, 10, 11, 12}, centroid = 7.

Final C_2 = {20, 25, 30}, centroid = 25.

2) Design of a clustering-based hospital care system

Goal: segment patients into clinically meaningful utilization and risk profiles so that a hospital can plan preventive care. The clusters are decision-support groups, *not* diagnoses.

- Data: age *band*, BMI, blood pressure, glucose, chronic-condition counts, medication count, emergency visits, admissions, missed appointments, laboratory summaries and activity measures. Personally identifying fields *are* excluded from modelling.
- Preprocessing: validate *units*, impute missing values, cap implausible outliers, one-hot encode categories, transform skewed visit counts and standardize numerical features. Feature *windows* must refer only to information available before the intervention date.
- Modelling: begin with K-Means for transparent centroid profiles. Compare Gaussian mixtures for overlapping groups and hierarchical clustering for nested structure. Run several initializations.

Q6 (cont.)

- Model selection: compare silhouette score, within-cluster dispersion, stability under resampling and review by clinicians. A mathematically compact cluster is useful only when its profile is *understandable* and actionable.
- Possible outputs: stable low-use *patients*; chronic-condition patients needing regular monitoring; high emergency-use patients; missed-appointment or adherence-risk patients; and recently deteriorating patients.
- Action layer: assign tailored reminders, care-manager calls, telehealth monitoring, nutrition support or medication review. Measure outcomes through controlled evaluation rather than *assuming* the clusters cause improvement.
- Governance: protect privacy, audit subgroup performance, avoid stigmatizing labels, allow clinical override and retrain when population or care pathways *change*.

The deployed system should show a patient profile and the nearest cluster centroid, together with uncertainty and the date of the last *model* update. This keeps the clustering result *interpretable* and supports responsible human decisions.

Q7 – Hard voting, soft voting and stacked generalization

1) Hard voting

Each base classifier outputs only a class label. The ensemble predicts the class receiving the largest number of votes. Weights may be added when some *models* are known to be more reliable. Hard voting works even when a model cannot produce probabilities, but it ignores confidence.

Hard-vote prediction = mode {h1(x), h2(x), ..., hm(x)}

2) Soft voting

Each classifier outputs a probability for every class. The probabilities are averaged, or combined by a weighted average, and the class with the largest combined probability is selected.

*Soft score for class k = sum_j wj * Pj(class k given x), with sum_j wj = 1.*

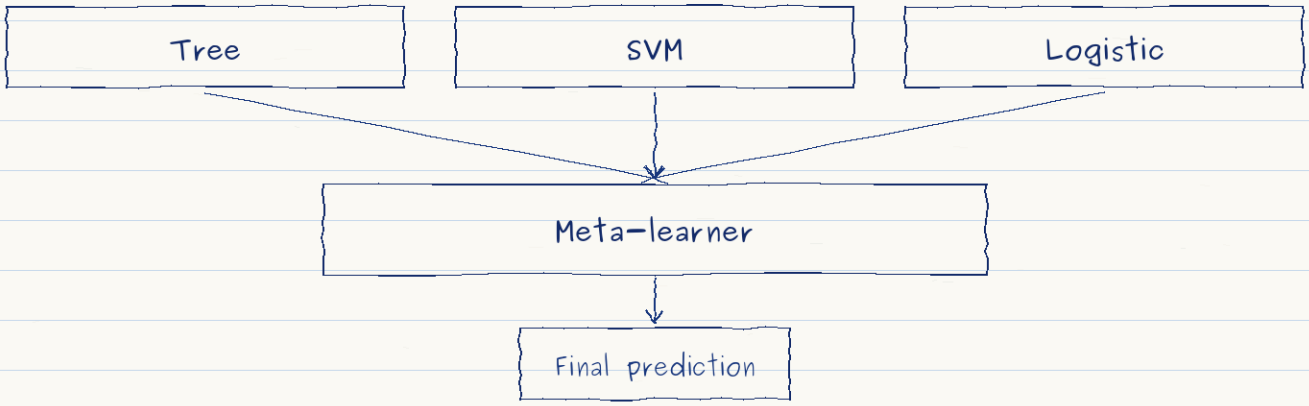
Soft voting uses *confidence and* can distinguish a weak vote from a very confident vote. It is most reliable when probabilities are reasonably *calibrated.* An overconfident, *poorly* calibrated model can *dominate* the average, so validation-based weighting or probability calibration may be *necessary.*

Small example

Suppose two models weakly prefer A with *probabilities* 0.51 and 0.55, while a third strongly prefers B with *probability* 0.99. Hard voting chooses A because A receives two labels. Soft voting can choose B because the combined probability for B is *larger.* Thus the two methods need not agree.

3) How stacking can improve accuracy

Q7 (cont.)



Stacked generalization trains several diverse base models *and* then trains a meta-learner on their predictions. The essential training data for the meta-learner are out-of-fold predictions: each base prediction for a training row must come from a model that did not train on that row. This prevents *target* leakage.

- A tree may be best for interaction-heavy cases, a linear model may be best near a broad linear trend, and another model may capture local structure.
- The meta-learner can learn conditional trust rather than using one fixed average. It may also correct systematic biases in base probabilities.
- Accuracy *improves* when base errors are diverse and the *validation* design matches *future* data.
- Improvement is not guaranteed. Highly correlated models, a small data set, leakage, or an overly complex meta-model can make stacking no better or even worse than the best individual model.
- Final evaluation must use a completely untouched test set after *all* base and meta-model *choices* are fixed.

Q8 – OR alternative: Random Forest, GBM and Stacking

1) Random Forest compared with Gradient Boosting Machines

Aspect	Random Forest	Gradient Boosting
Construction	Many trees on bootstrap samples; random feature subsets	Trees added sequentially to correct residual or gradient error
Main effect	Strong variance reduction	Strong bias reduction, with variance controlled by tuning
Training	Trees can be trained in parallel	Sequential; later trees depend on earlier trees
Tuning	Usually robust with reasonable defaults	Sensitive to depth, learning rate, tree count and regularization
Noise	Generally robust to noise and outliers	Can chase noise if boosted too long or too deeply
Speed	Often faster and easier to parallelize	Usually slower; careful validation is needed
Accuracy	Excellent dependable baseline	Often higher on structured/tabular data when well tuned

Random Forest – advantages

- Reduces the instability and overfitting of a single deep tree by averaging many decorrelated trees.
- Handles nonlinear relationships and interactions with little preprocessing and no feature scaling.
- Supports *out-of-bag* error estimation and feature-importance summaries.
- Parallel training and broad robustness make it a strong first choice.

Random Forest – limitations

- A large forest uses memory and is *less* interpretable than one tree.
- Averaging mainly reduces variance; it may not remove a strong bias from weak feature representation.
- Standard impurity importance can favour high-cardinality or highly variable features. Permutation importance is safer *for* interpretation.
- *Predictions* are piecewise averages and may extrapolate poorly in regression.

Gradient Boosting – advantages

- Sequential error correction can achieve very high predictive accuracy.
- A small learning rate with shallow trees models complex nonlinear patterns while regularizing each step.

Q8 (cont.)

- Modern implementations can handle missing values, regularization and efficient histogram splits.

Gradient Boosting – limitations

- More hyperparameters and stronger interaction among them make tuning important.
- Training is sequential and therefore less parallel than a forest.
- Deep trees, a high learning rate or too many rounds can overfit noise.
- Results can be less stable under distribution shift, so *early* stopping and careful validation are essential.

Use Random Forest when a robust, *low-tuning* baseline and parallel training are priorities. Use Gradient Boosting when maximum tabular-data *accuracy* justifies tuning and latency. The correct choice should be made by validation under the real deployment metric and *constraints*.

2) Stacked Generalization (Stacking)

Stacking is *an* ensemble architecture in which a second-level model learns how to combine the outputs of first-level models.

- Base learners: diverse algorithms such as trees, linear models, SVMs or neural networks.
- Level-one features: class probabilities or numerical predictions *produced* by the base learners. Original input features may also be included.
- Meta-learner: a usually simple model, such *as* regularized Logistic Regression for classification or linear regression for regression.
- *Cross-validation* mechanism: generates out-of-fold predictions *so* the meta-learner never trains on in-sample base predictions.
- Final refit: after the meta-learner design is fixed, each base learner is trained on all training data. At inference, their outputs are passed to the meta-learner.

Training procedure

- Split the training data into K folds.
- For each base model and each fold, train on K−1 folds and predict the held-out fold.
- Join the *held-out* predictions to form a complete level-one training matrix.
- Train the meta-learner on this matrix and the true targets.

Q8 (cont.)

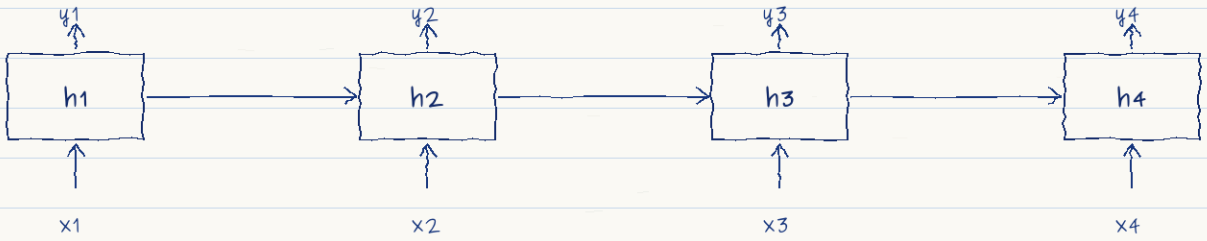
- Refit base models on all training data, then evaluate the complete stack once on an untouched test set.

Stacking can *outperform* each component when their errors are complementary. It also adds computational cost and a serious risk of leakage *if* ordinary in-sample predictions are used for *meta-training*.

Q9 – Recurrent Neural Networks and activation functions

1) RNN architecture

A Recurrent Neural Network is designed for sequential data. Unlike an MLP that maps a fixed input vector directly to an output, an RNN maintains a hidden state *that* carries information from earlier time steps. The same parameters are reused at every position in the sequence.



Same weights are reused at every time step.

$$h(t) = phi[\ Wxh * x(t) + Whh * h(t-1) + bh\ ]$$

$$y(t) = g[\ Why * h(t) + by\ ]$$

Here x(t) is the current input, h(t−1) is the previous memory state, phi is the hidden activation, and g is the output activation. Unrolling the network through time shows a chain of identical recurrent cells. Because weights are shared, the model can process variable-length sequences and detect a pattern *regardless* of its absolute position.

Sequence arrangements

- Many-to-one: a whole sequence gives one output, for example sentiment classification.
- One-to-many: one context produces a sequence, for example basic sequence generation.
- Many-to-many aligned: *an* output is *produced* at each time *step*, for example sequence labelling.
- Encoder-decoder: an input sequence is summarized and used *to* generate an output sequence.

How time dependence is handled

Q9 (cont.)

The hidden state acts as a running summary. A current prediction depends on both the present observation and the state created by earlier observations. During training, backpropagation through time unfolds the recurrence and accumulates gradients across time steps. The order of observations is therefore part of the model, not discarded.

Vanilla RNNs can suffer vanishing gradients, which makes long-range dependencies hard to learn, and exploding gradients, which cause unstable updates. Gradient clipping limits exploding gradients. LSTM and GRU cells introduce gates and additive memory paths, allowing information to be retained or forgotten more selectively.

Applications

- Language modelling, text classification and sequence tagging.
- Speech recognition and audio-event modelling.
- Time-series forecasting for demand, finance, weather or sensors.
- Anomaly detection in machine, network or medical streams.
- Video-frame and human-activity sequence analysis.

RNNs are useful when sequence order and recent context are important, especially for streaming or moderate-length sequences. For very long contexts and highly parallel training, attention-based Transformers may be more effective, but the recurrent state remains valuable in compact online systems.

2) Critical comparison of activation functions

Sigmoid: sigma(z) = 1 / (1 + exp(−z))

- Range 0 to 1, so it has a clear probability interpretation at a binary output.
- Smooth and differentiable, and useful as a gate in LSTM or GRU units.
- For large positive or negative z it saturates. The derivative becomes very small, causing vanishing gradients.
- Outputs are not zero-centred, which can slow hidden-layer optimization.
- Best use: binary-classification output or gates, not usually deep hidden layers.

Tanh: tanh(z)

- Range −1 to 1 and zero-centred, so it is often preferable to sigmoid for a hidden state.

Q9 (cont.)

- Provides both positive and negative activations and is a traditional *RNN* hidden activation.
- Still saturates for large absolute inputs and therefore still *suffers* vanishing gradients.
- Best use: bounded hidden states in small networks or recurrent cells, especially when inputs are normalized.

ReLU: max(0, z)

- Very inexpensive and has gradient 1 on the positive side, reducing the positive-side saturation problem.
- Produces sparse activations because all negative inputs become zero.
- A unit can die if it remains *on* the negative side, where the gradient is zero.
- Output is unbounded, so initialization, learning rate and normalization matter.
- Best use: default hidden activation for many MLP and convolutional layers.

Leaky ReLU: max(alpha*z, z), with small alpha > 0

- Retains a small negative-side gradient, greatly reducing the *dying-ReLU* problem.
- Keeps much of ReLU efficiency and *positive-side* behaviour.
- The negative slope is an extra design choice *and* activations are less sparse than ReLU.
- Best use: hidden layers when ordinary ReLU units die or when negative *information* should not be removed completely.

Layer choice summary: hidden layers -> ReLU or Leaky ReLU; binary output -> Sigmoid; bounded recurrent state -> often Tanh.

No activation is universally best. Suitability depends on the layer role, initialization, normalization, optimization method and the required output range.

Q10 – OR alternative: recurrent networks and optimization

1) When to use recurrent networks instead of an MLP

Use a recurrent network when the input is an ordered sequence *and* the meaning of the current element depends on previous elements. An MLP treats its input as one fixed-size vector and has no built-in memory or shared operation across time.

- Choose an RNN, LSTM or GRU for speech, text, sensor streams, event logs and time series where order matters.
- Use recurrence when sequence lengths vary and a running state *must* be updated as new observations arrive.
- Use it when *temporal* patterns such as delay, trend, periodicity or context affect the prediction.
- An MLP is usually sufficient when observations are independent, features have a fixed meaning and sequence order adds no information.
- For a short fixed window, lagged values can be flattened into an MLP, but this does *not* share parameters naturally across time.
- For very long-range dependencies or maximum parallelism, compare RNNs with temporal CNNs and Transformers rather than assuming recurrence is best.

The final judgment depends on *validation* accuracy, *sequence* length, latency, memory and interpretability. A simpler MLP should be preferred when it performs equally well on a non-sequential representation.

2) Gradient Descent versus Stochastic Gradient Descent

*Batch GD: theta <- theta − eta * (1/N) * sum_i gradient L_i(theta)*

*SGD: theta <- theta − eta * gradient L_i(theta)*

Property	Gradient Descent	Stochastic Gradient Descent
Data per update	Entire training set	One randomly selected example
Update direction	Accurate and deterministic	Noisy estimate of full gradient
Cost per update	High for large N	Very low
Convergence path	Smooth, stable	Fluctuates around the optimum
Best setting	Small or moderate data	Large or streaming data
Parallel hardware	Can use matrix operations	Pure one-sample updates underuse hardware

Q10 (cont.)

Batch Gradient Descent computes the exact gradient of the empirical loss before every update. It is stable but each step becomes expensive when the data set is large. SGD updates after one example, so it begins learning quickly and its noise can help move through flat regions, but the learning rate normally has to decrease for convergence.

In practice, mini-batch SGD is the usual compromise: it uses a small batch, gives an efficient vectorized update, and has enough noise for effective optimization. Momentum, Adam or learning-rate schedules are often added. A final model must be chosen using validation performance rather than the apparent smoothness of the training loss.